Olist Brazilian E-commerce – Data Analytics Capstone

Executive Summary

Project Overview

This project analyzes the Olist Brazilian e-commerce dataset to understand business performance, delivery operations, customer satisfaction, revenue patterns and payment behavior. The objective is to identify operational challenges and provide data-driven recommendations to improve business performance and customer experience.

Business Problem

The central business question was: **Are delivery delays affecting customer satisfaction, and what operational improvements should Olist prioritize?**

Five areas were investigated:

- The relationship between delivery delays and customer review scores.
- Geographical differences in delivery performance.
- Revenue contribution by product category and seller.
- Payment method popularity and payment values.
- The most important operational improvement area.

Tools and Technologies

- **Python:** Pandas, NumPy, Matplotlib, Seaborn and SciPy for data cleaning, analysis and statistical testing.
- **SQL:** PostgreSQL for querying, joining and analyzing structured data using joins, CTEs and window functions.
- **Power BI:** Data modeling, DAX measures and interactive dashboard development.
- **Google Colab:** Python notebook environment.

Key Findings

- The dataset contains **99,441 orders** and multiple related datasets covering customers, products, sellers, payments and reviews.
- Approximately **97.02% of orders were delivered**.
- **7,826 delivered orders were late**, representing a late-delivery rate of **8.11%**.

- The overall average customer review score was **4.05 out of 5**.
- Approximately **14.70% of review records were low ratings (1–2)**.
- Among reviewed late deliveries, **54.02% received low ratings**, compared with **9.19% of reviewed on-time deliveries**.

Statistical Analysis

A Chi-square test of independence was conducted to examine the relationship between delivery status and customer review categories.

The test produced a Chi-square statistic of **13,249.786**, with **2 degrees of freedom** and a **p-value below 0.001**. The null hypothesis of independence was rejected, indicating a statistically significant association between delivery status and review ratings.

This result demonstrates an association, not proof of causation.

Business Recommendations

1. Prioritize delivery reliability and reduce late deliveries.
2. Monitor delivery performance across geographical regions and identify areas with recurring delays.
3. Evaluate seller-level delivery performance and support improvements in shipping operations.
4. Improve delivery estimates and proactively communicate delays to customers.
5. Monitor revenue contribution by category and seller, alongside payment behavior and customer satisfaction.

Conclusion

The analysis identifies delivery reliability as the highest-priority operational improvement area. Although most orders were delivered, late deliveries were strongly associated with lower customer ratings. By improving delivery monitoring, seller coordination and customer communication, Olist can work toward a more reliable delivery experience and stronger customer satisfaction.

Project Outcome: A complete analytical workflow covering data exploration, cleaning, modeling, SQL analysis, Python statistical analysis, Power BI dashboard development and business recommendations.